3 Proof of the main results

3.1 Annealed free energy and coefficient asymptotics

To identify the deterministic coefficient $A_N(\beta)$ in (**??**), we use the annealed free-energy limit from [**?**]. The required convergence of derivatives directly follows from its convexity.

The following is a standard convergence theorem for convex functions; see [**?**, Theorems 10.8 and 25.7].

Lemma 3.1. *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and that $h_N \to h$ pointwise, where $h \in C^1(I)$. Then locally uniformly on I,*

$$h_N \to h \qquad \text{and} \quad h_N' \to h'.$$

Proof. Fix $[a, b] \subset I$. For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x-s)}{s} \le h_N'(x) \le \frac{h_N(x+s) - h_N(x)}{s}.$$

Taking lim inf and lim sup as $N \to \infty$, using pointwise convergence, and then letting $s \downarrow 0$, gives

$$h'(x) \le \liminf_N h_N'(x) \le \limsup_N h_N'(x) \le h'(x).$$

Thus $h_N'(x) \to h'(x)$ pointwise.

Each h_N' is nondecreasing. Given $\varepsilon > 0$, by uniform continuity of h' on $[a, b]$, choose a partition $a = x_0 < x_1 < \cdots < x_m = b$ such that, for every $i = 0, \dots, m-1$,

$$\sup_{x,y \in [x_i, x_{i+1}]} |h'(x) - h'(y)| < \varepsilon.$$

Pointwise convergence at the finitely many x_i gives, for all sufficiently large N and every $i = 0, \dots, m$,

$$|h_N'(x_i) - h'(x_i)| < \varepsilon.$$

For $x \in [x_i, x_{i+1}]$, monotonicity of h_N' gives

$$h_N'(x_i) \le h_N'(x) \le h_N'(x_{i+1}),$$

and hence

$$|h_N'(x) - h'(x)| < 2\varepsilon.$$

Therefore, $h_N' \to h'$ uniformly on $[a, b]$.

By the mean value theorem, we also have that $h_N \to h$ uniformly on $[a, b]$. □

Proposition 3.2. *There are $\beta_{\mathrm{FE}} = \beta_{\mathrm{FE}}(q) > 0$ and an even real-analytic function ℓ_q on $(-\beta_{\mathrm{FE}}, \beta_{\mathrm{FE}})$ such that, locally uniformly on this interval,*

$$\frac{1}{N} \log \mathbb{E} Z_N(\beta) \longrightarrow \ell_q(\beta), \tag{1}$$

$$\frac{1}{N} \partial_\beta \log \mathbb{E} Z_N(\beta) \longrightarrow \ell_q'(\beta). \tag{2}$$

Moreover,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3). \tag{3}$$

In paritcular, $\ell'(\beta) > 0$, for every sufficiently small real $\beta \neq 0$.

Proof. The pointwise convergence in (1) follows from Theorem 5 of [**?**], together with the universality [**?**]. To make it more precise, we record the normalization conversion. The Majoranas of [**?**] satisfy

$$\psi_i^{\mathrm{GPSZ}} = 2^{-1/2}\psi_i, \qquad H_N^{\mathrm{GPSZ}} = 2^{-q/2}H_N.$$

Set $u := 2^{q/2}\beta$ so that $uH_N^{\mathrm{GPSZ}} = \beta H_N$. Thus the normalization conventions of [**?**] give

$$\frac{1}{N}\log \mathbb{E}Z_N(\beta) = \frac{u}{N} f_{N,q}^{\mathrm{GPSZ}}(u), \tag{4}$$

where $f_{N,q}^{\mathrm{GPSZ}}(u) = u^{-1}\log \mathbb{E}Z_N^{\mathrm{GPSZ}}(u)$.

We next extract the regularity and the first coefficient from the explicit formula in that theorem. In its notation,

$$\Phi_u(z) = \sum_{v\geq 1} \mu_v(u) z^v, \qquad u f_q^{\mathrm{GPSZ}}(u) = \Phi_u(t_*(u)) - 1 - \log t_*(u),$$

where $t_*\partial_z\Phi_u(t_*) = 1$. A contributing hypertree has $v = 1 + (q-1)s$ vertices and its coefficient contains

$$\rho(u)^{v-1}, \qquad \rho(u)^{q-1} = \frac{(q-1)!u^2}{2^{q+1}}.$$

More explicitly, $\mathcal{T}_v^{\mathrm{sim}}$ is empty unless $v = 1 + (q-1)s$ for some $s \geq 0$. For $s \geq 1$ we may therefore write

$$\mu_{1+(q-1)s}(u) = c_{q,s}u^{2s},$$

where

$$c_{q,s} = \frac{1}{(1+(q-1)s)!}\left(\frac{(q-1)!}{2^{q+1}}\right)^s \sum_{T\in\mathcal{T}_{1+(q-1)s}^{\mathrm{sim}}} b_T.$$

The bounds used in the proof of Proposition 3 of [**?**] give $|c_{q,s}| \leq C_q^s$. Hence

$$\Phi_u(z) = z + \sum_{s\geq 1} c_{q,s}u^{2s}z^{1+(q-1)s}$$

converges normally for (u^2, z) in a neighborhood of $(0,1)$. Thus $\Phi_u(z)$ is jointly analytic in (u^2, z) there. At $u = 0$, $\Phi_0(z) = z$ and $t_*(0) = 1$; the derivative with respect to t of $t\partial_z\Phi_u(t) - 1$ is 1 at $(u,t) = (0,1)$. The analytic implicit function theorem makes t_*, and hence the limiting right-hand side, analytic in u^2.

The first nontrivial term is the unique one-edge hypertree on q vertices:

$$\Phi_u(z) = z + \frac{u^2}{q2^{q+1}}z^q + O(u^4).$$

It follows successively that

$$t_*(u) = 1 - \frac{u^2}{2^{q+1}} + O(u^4), \qquad u f_q^{\mathrm{GPSZ}}(u) = \frac{u^2}{q2^{q+1}} + O(u^4).$$

Using $u = 2^{q/2}\beta$ in (4) gives (3).

It remains only to justify the two local-uniform assertions. Let $h_N := N^{-1}\log F_N$. If

$$\overline{\nu}_N(B) = \mathbb{E}\ \mathrm{tr}\,\mathbf{1}_{\{H_N\in B\}},$$

then $\mathbb{E}Z_N(\beta) = \int e^{-\beta x}\overline{\nu}_N(\mathrm{d}x)$; hence h_N is finite, differentiable, and convex. Differentiation is justified by $\|H_N(G)\| \leq \sigma_N \sum_{I\in\mathcal{I}_N}|g_I|$ and Gaussian exponential integrability. The limiting function just obtained is C^1. Lemma 3.1 now proves both local-uniform convergences. □

By (**??**) and (2),

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell'_q(\beta).$$

Proof of Theorem **??**. Let

$$S_N := \frac{1}{\sqrt{2M_N}} \sum_{I\in\mathcal{I}_N} (g_I^2 - 1).$$

The classical i.i.d. central limit theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma **??**,

$$Y_N(\beta) - 1 = c_N(\beta)S_N + \mathcal{R}_N(\beta), \qquad c_N(\beta) := \frac{\beta\,\partial_\beta \log \mathbb{E}Z_N(\beta)}{\sqrt{2M_N}}, \tag{5}$$

where $\mathcal{R}_N := \sum_{r\geq 3} P_r Y_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (2) gives

$$N^{q/2-1}c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell'_q(\beta).$$

Proposition **??** gives $N^{q/2-1}\mathcal{R}_N \to 0$ in L^2. This proves (**??**). Orthogonality in (5) also gives

$$\mathrm{Var}(Y_N) = c_N(\beta)^2 + \|\mathcal{R}_N(\beta)\|_2^2\,.$$

Finally, $\mathrm{Var}(Y_N) = \mathrm{Var}(Z_N)/(\mathbb{E}Z_N)^2$, and $\ell'(\beta) > 0$ for β small enough give (**??**). □

[1]

Proof of Corollary **??**. The case $\beta = 0$ is trivial, so assume $\beta \neq 0$. Set

$$a := \frac{q}{2} - 1, \qquad W_N := \log Y_N(\beta).$$

Then

$$N^a\big(\log Z_N - \mathbb{E}\log Z_N\big) = N^a\big(W_N - (Y_N - 1)\big) + N^a(Y_N - 1) - N^a\mathbb{E}W_N.$$

By Theorem **??**,

$$N^a(Y_N - 1) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell'_q(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1).$$

Moreover, $Y_N - 1 \to 0$ in probability and $N^a(Y_N - 1)$ is tight. Since whenever $|x| \leq 1/2$,

$$|\log(1+x) - x| \leq x^2,$$

it follows that

$$N^a\big(W_N - (Y_N - 1)\big) \xrightarrow{\mathrm{p}} 0.$$

Thus it remains to prove

$$\mathbb{E}W_N = o(N^{-a}).$$

Write, as in (5),

$$Y_N - 1 = c_N S_N + \mathcal{R}_N,$$

where

$$c_N = \frac{\beta\,\partial_\beta \log \mathbb{E}Z_N(\beta)}{\sqrt{2M_N}} = O(N^{-a}), \quad S_N = \frac{1}{\sqrt{2M_N}} \sum_{I\in\mathcal{I}_N} (g_I^2 - 1), \quad \|\mathcal{R}_N\|_2^2 = O(N^{-3a})$$

[1] SN: The following is a revised proof

by Proposition **??**. We note that

$$\sup_N \mathbb{E}S_N^4 < \infty.$$

Orthogonality therefore gives

$$\mathbb{E}(Y_N - 1)^2 = O(N^{-2a}).$$

On the other hand, we estimate

$$\begin{aligned}\mathbb{P}(Y_N < 1/2) &\le \mathbb{P}(|c_N S_N| > 1/4) + \mathbb{P}(|\mathcal{R}_N| > 1/4)\\ &\le Cc_N^4 \mathbb{E}S_N^4 + C\|\mathcal{R}_N\|_2^2\\ &= O(N^{-3a}).\end{aligned}$$

Next we will prove that

$$Z_N(\beta; G) \ge 1.$$

Choose $x \in I$. The unitary U_x from Lemma **??** satisfies

$$U_x \psi_I U_x^* = -\psi_I,$$

and hence $\operatorname{tr}\psi_I = 0$. Therefore $\operatorname{tr} H_N(G) = 0$. Jensen's inequality applied to the eigenvalues of $H_N(G)$ now gives

$$Z_N(\beta; G) = \operatorname{tr} e^{-\beta H_N(G)} \ge e^{-\beta \operatorname{tr} H_N(G)} = 1.$$

Define

$$h(y) := y - 1 - \log y, \qquad y > 0.$$

Since $\mathbb{E}Y_N = 1$,

$$-\mathbb{E}W_N = \mathbb{E}h(Y_N).$$

On $\{Y_N \ge 1/2\}$,

$$h(Y_N) \le (Y_N - 1)^2,$$

and therefore

$$\mathbb{E}\big[h(Y_N)\mathbf{1}_{\{Y_N \ge 1/2\}}\big] = O(N^{-2a}).$$

On the other hand, if $Y_N < 1/2$, then

$$h(Y_N) \le -\log Y_N.$$

Since $Z_N \ge 1$,

$$-\log Y_N = \log \mathbb{E}Z_N - \log Z_N \le \log \mathbb{E}Z_N.$$

By Proposition 3.2, $\log \mathbb{E}Z_N = O(N)$, we have

$$\mathbb{E}\big[h(Y_N)\mathbf{1}_{\{Y_N < 1/2\}}\big] \le O(N)\,\mathbb{P}(Y_N < 1/2) = O(N^{1-3a}).$$

Since $q \ge 4$, we have $a \ge 1$, and hence $1 - 3a \le -2a$. Consequently,

$$0 \le -\mathbb{E}W_N = O(N^{-2a}) = O(N^{2-q}) = o(N^{-a}).$$

Combining the three terms above and applying Slutsky's theorem yields

$$N^{q/2-1}\big(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\big) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\beta^2(\ell_q'(\beta))^2\right).$$

□

Lemma 3.3. [2] *Write*

$$W_N := \log Y_N(\beta).$$

Then, for $\beta \neq 0$ and any $\lambda, t > 0$,

$$\mathbb{P}(W_N - \mathbb{E}W_N < -t) \leq \exp\left(-\frac{qt^2}{2\beta^2}\right), \tag{6}$$

$$\mathbb{E}\exp(\lambda(W_N - \mathbb{E}W_N)) \leq \exp\left(\frac{\lambda^2\beta^2}{2q}\right). \tag{7}$$

Proof. To apply the Gaussian concentration inequality, we view $Z_N(\beta; G)$ as a function of $G := (g_I)_{I\in\mathcal{I}_N}$ and establish the Lipschitz bound for W_N. Fix $I \in \mathcal{I}_N$. By the one-variable Duhamel formula (**??**), we deduce that

$$\begin{aligned}
\partial_{g_I} Z_N &= \operatorname{tr} \partial_{g_I} e^{-\beta H_N(G)} \\
&= -\beta \operatorname{tr}\Big(\partial_{g_I} H_N(G) \cdot e^{-\beta H_N(G)}\Big) \\
&= -\beta\sigma_N \operatorname{tr}\big(e^{-\beta H_N}\psi_I\big).
\end{aligned}$$

Write $\langle\cdot\rangle_G$ for the expectation with respect to the normalized Gibbs state, i.e.,

$$\langle A\rangle_G := \frac{\operatorname{tr}(Ae^{-\beta H_N(G)})}{Z_N(\beta; G)}.$$

Then we obtain

$$\partial_{g_I} W_N = -\beta\sigma_N\langle\psi_I\rangle_G. \tag{8}$$

Fix an arbitrary subset $S \subseteq [N]$ with $|S| = q - 1$. Set $A_i := \psi_{S\cup\{i\}}$ for each $i \notin S$. Note that, by the relation $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$, it holds that

$$A_i^2 = \mathbb{I}, \qquad A_iA_j + A_jA_i = 0 \quad \text{if} \quad i \neq j. \tag{9}$$

This yields

$$\left(\sum_{i\notin S}\langle A_i\rangle_G A_i\right)^2 = \sum_{i\notin S}\langle A_i\rangle_G^2 A_i^2 + \sum_{\substack{i,j\notin S\\ i<j}}\langle A_i\rangle_G\langle A_j\rangle_G(A_iA_j + A_jA_i) = \sum_{i\notin S}\langle A_i\rangle_G^2\mathbb{I}.$$

If we write $\|\cdot\|$ for the operator norm, then, by (9), we obtain

$$\left\|\left(\sum_{i\notin S}\langle A_i\rangle_G A_i\right)^2\right\| = \sum_{i\notin S}\langle A_i\rangle_G^2.$$

Since A_i is self-adjoint, it follows that

$$\left\|\sum_{i\notin S}\langle A_i\rangle_G A_i\right\| = \sqrt{\sum_{i\notin S}\langle A_i\rangle_G^2}. \tag{10}$$

On the other hand, Hölder's inequality for Schatten norms yields

$$\left\langle\sum_{i\notin S}\langle A_i\rangle_G A_i\right\rangle_G \leq \left\|\sum_{i\notin S}\langle A_i\rangle_G A_i\right\|. \tag{11}$$

[2] SN: I think we can remove the blue.

Since the left-hand side equals $\sum_{i\notin S}\langle A_i\rangle_G^2$, we deduce from (10) and (11) that

$$\sum_{i\notin S}\langle A_i\rangle^2 \le 1. \tag{12}$$

For each $I \in \mathcal{I}_N$, there exist exactly q subsets $S \subseteq [N]$ satisfying $I = S \cup \{i\}$. Hence,

$$\sum_{\substack{S\subseteq[N]\\|S|=q-1}}\sum_{i\notin S}\langle\psi_{S\cup\{i\}}\rangle_G^2 = q\sum_{I\in\mathcal{I}_N}\langle\psi_I\rangle_G^2.$$

This, combined with (12), yields

$$\sum_{I\in\mathcal{I}_N}\langle\psi_I\rangle_G^2 \le \frac{1}{q}\sum_{\substack{S\subseteq[N]\\|S|=q-1}} 1 = \frac{1}{q}\binom{N}{q-1} \le \frac{N^{q-1}}{q!}.$$

From this and (8), we deduce that

$$\|\nabla W_N\|_2^2 \le \frac{\beta^2\sigma_N^2 N^{q-1}}{q!} = \frac{\beta^2}{q},$$

where the last equality follows from the definition of σ_N^2. This implies that $W_N(G)$ is $|\beta|/\sqrt{q}$-Lipschitz as a function of $G = (g_I)_{I\in\mathcal{I}_N}$. Hence, the Gaussian concentration inequality gives the desired result. □

Proof of Corollary **??**. The case $\beta = 0$ is trivial, so assume $\beta \neq 0$. Put

$$a := \frac{q}{2} - 1, \qquad W_N := \log Y_N(\beta).$$

It then holds that

$$\begin{aligned}\log Z_N - \mathbb{E}\log Z_N &= (\log Z_N - \log \mathbb{E}Z_N) + (\log \mathbb{E}Z_N - \mathbb{E}\log Z_N)\\ &= W_N - \mathbb{E}W_N\\ &= \big(W_N - (Y_N(\beta)-1)\big) + (Y_N(\beta)-1) - \mathbb{E}W_N,\end{aligned}$$

which yields

$$N^a(\log Z_N - \mathbb{E}\log Z_N) = N^a\big(W_N - (Y_N(\beta)-1)\big) + N^a(Y_N(\beta)-1) - N^a\mathbb{E}W_N.$$

By Theorem **??**, we have

$$N^a(Y_N(\beta)-1) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi. \tag{13}$$

Let $C_0 \in (0,\infty)$ be such that $|\log(1+x)-x| \le C_0x^2$ for all $x \in [-1/2, 1/2]$. Since $W_N = \log Y_N(\beta) = \log(1+(Y_N(\beta)-1))$, it follows that

$$\mathbb{E}\big[1\wedge(N^a|W_N-(Y_N(\beta)-1)|)\big] \le \mathbb{E}\big[1\wedge(C_0N^a|Y_N(\beta)-1|^2)\big] + \mathbb{P}(|Y_N(\beta)-1|>1/2).$$

By (13), $|Y_N(\beta)-1| \xrightarrow{\mathrm{p}} 0$ and $(N^a|Y_N(\beta)-1|)_{N\ge1}$ is tight. Thus, the right-hand side of the above inequality converges to 0 as $N\to\infty$. Hence, $N^a(W_N-(Y_N(\beta)-1)) \xrightarrow{\mathrm{p}} 0$. Therefore, it suffices to show that $\mathbb{E}W_N = o(N^{-a})$.

We first claim that

$$-\frac{\beta^2}{2q} \le \mathbb{E}W_N \le 0. \tag{14}$$

Indeed, the upper bound follows from Jensen's inequality, while the lower bound follows from (7) with $\lambda = 1$ and the identity $\mathbb{E}e^{W_N} = 1$. By Theorem **??**,

$$\delta_N^2 := \mathbb{E}(Y_N(\beta) - 1)^2 = O(N^{-(q-2)}). \tag{15}$$

Henceforth, we consider sufficiently large N so that $\delta_N \le e^{-1}$. Since $e^{W_N} - 1 = Y_N(\beta) - 1$, Markov's inequality implies that for any $t \ge 1$,

$$\mathbb{P}(W_N < -t) \le \frac{\delta_N^2}{(1 - e^{-t})^2} \le C\delta_N^2. \tag{16}$$

On the other hand, by $-\mathbb{E}W_N \le K_\beta := \beta^2/(2q)$ due to (14) and (6), for any $t > K_\beta$,

$$\mathbb{P}(W_N < -t) \le \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right). \tag{17}$$

Set

$$T_N := \max\left\{1,\, K_\beta + \frac{2|\beta|}{\sqrt{q}}\sqrt{\log(1/\delta_N)}\right\}.$$

We represent

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] = \mathbb{P}(W_N < -1) + \int_1^\infty \mathbb{P}(W_N < -t)\,\mathrm{d}t.$$

Applying (16) to the first term and the integral over $[1, T_N]$ and applying (17) to the integral over (T_N, ∞), we obtain

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le CT_N\delta_N^2 + \int_{T_N}^\infty \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right)\mathrm{d}t.$$

By the Gaussian tail bound $\int_a^\infty e^{-t^2}\mathrm{d}t \le Ce^{-a^2}$ for any $a \ge 0$, the second term is bounded by $C\delta_N^2$. Consequently,

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}. \tag{18}$$

Now set $h(w) := e^w - 1 - w$. Note that

$$h(w) \le C(e^w - 1)^2 \quad (w \ge -1), \qquad h(w) \le |w| \quad (w < -1).$$

This, combined with $e^{W_N} - 1 = Y_N(\beta) - 1$ and (18), yields

$$\mathbb{E}h(W_N) \le C\mathbb{E}(Y_N(\beta) - 1)^2 + \mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}.$$

Moreover, since $h(W_N) = Y_N(\beta) - 1 - W_N$ and $\mathbb{E}(Y_N(\beta) - 1) = 0$, we have $-\mathbb{E}W_N = \mathbb{E}h(W_N)$. Therefore, by (14), we deduce that

$$0 \le -\mathbb{E}W_N \le C\delta_N^2\sqrt{\log(1/\delta_N)}.$$

From this and (15), it follows that

$$\mathbb{E}W_N = O\Big(N^{2-q}\sqrt{\log N}\Big) = o(N^{-(q/2-1)}),$$

which completes the proof. □